

Fourth Semester Diploma Examination in Automobile Engineering.**Model Question Paper.****Automobile Electrical and Electronic Systems**

Maximum marks: 100.

Time: 3 hours.

Part A

(Answer the following questions in one or two sentences. Each question carries 2 marks.)

I

1. Name 2 types of materials used as separators in lead acid cell
2. List the 2 important parts of an alternator using in an automobile
3. State the use of ballast resistor in ignition system
4. Define Dwell Angle
5. Recognize two anti-dazzling methods

Part B

(Answer any five of the following. Each question carries 6 marks.)

II

1. Summarize the steps for conducting high discharge test and sketch a cell tester
2. Explain the working of a solenoid switch in starting system
3. Conclude the temperature effect of voltage regulator
4. Describe the constructional details of a Distributor
5. Differentiate between magneto ignition system and battery coil ignition system
6. Indicate the use of horn relay and explain its working
7. Illustrate "the autronic eye" with a diagram

Part C

(Answer one full question from each unit. Each question carries 15 marks.)

Module - I**III**

- a) Formulate the equation for charging and discharging of a nickel cadmium cell and describe its working. 8
- b) Identify three battery defects and explain about the causes of these defects. 7

OR

IV		
	a) Demonstrate constant voltage charging of battery with a connection diagram	8
	b) Differentiate Ampere hour efficiency and Watt hour efficiency of battery.	7
	Module -II	
V		
	a) Define the term armature reaction and justify its utilization for output control of generator.	7
	b) Recognize 4 generator complaints and remedies	8
	OR	
VI		
	a) Illustrate the working of a alternator with the help of a neat diagram	8
	b) Explain the constructional details of Bendix drive mechanism in starter motor, with sketch.	7
	Module -III	
VII		
	a) Identify the need of spark advance & retard mechanisms	8
	b) Label any one of the magneto ignition systems and conclude its working	7
	OR	
VIII		
	a) Describe capacitive discharge ignition system with a neat diagram	8
	b) Compare between can type and core type ignition coil	7
	Module -IV	
IX		
	a) Reproduce the wiring diagram of a two-wheeler and identify the parts	8
	b) Interpret the principle of operation of Oil Pressure Gauge	7
	OR	
X		
	a) Generate the arrangements to be performed for head light alignment	8
	b) Demonstrate the principle of operation of 3 speed wind screen wiper motor.	7